

Prime Dynamics Program, Volume IV

Noisy Heads, Annular Endpoints, First Alias, and Signed Completion

A Provenance-Preserving Synthesis of RH-282–RH-361

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Abstract

This fourth volume synthesizes eighty numbered papers, RH-282–RH-361, at the point where projection-free noisy spectral tails meet weighted prefixes, annular endpoint criteria, first-alias boundary layers, and deterministic counterloops. The range contains substantial exact mathematics: canonical finite noisy heads, regularized-determinant tail bounds, sharp clock and mass barriers, endpoint expansions, exact local affine interfaces, corrected trace observations, boundary-orbit decompositions, and an unconditional deterministic terminal-lag phase diagram. These statements live in several different data types and cannot be merged by notation alone.

We give a machine-checked provenance theorem for all eighty labels and the four historical duplicate-directory groups. We then state the exact common clock identities

$$p = \tau - a = q - d, \quad d = h - s, \quad q = p + d, \quad h = s + d,$$

where p, q, h are actual coefficients or heads in their declared physical types and s is the deterministic graded counterloop. A typed nonpromotion theorem shows that the selected or normalized actual branch and the deterministic branch do not determine an unnormalized full-trace or head budget without an independent theorem for $d = h - s$.

The first missing physical leaf is

$$D_{4k}(R) = \sum_{2 \leq n < 4k} |h_{\sigma,n} - s_{k,n}| \frac{R^n}{n} \longrightarrow 0.$$

No source in the volume proves this limit, proves a physical obstruction to it, or supplies an equivalent complete q/E_{off} theorem. The resulting verdict is source-relative NOT_TESTABLE, not nonexistence. Gates A–E remain false/open. This synthesis does not construct a Hilbert–Polya operator, identify Riemann zeros, prove a von Mangoldt trace, prove completed-zeta divisor equality, or prove the Riemann Hypothesis. RH-362 is not activated.

Keywords: transfer operators; regularized determinants; spectral heads; annular Hardy bounds; first alias; signed completion; provenance audit.

1 Series position and rules of synthesis

The four-volume partition is functional rather than chronological rhetoric: Volume I covers RH-1–RH-160, Volume II covers RH-161–RH-241, Volume III covers RH-242–RH-281, and this volume covers RH-282–RH-361. The original numbered papers remain the atomic sources. A review anchor records status and dependency information, but it does not replace the proof in the source paper it reviews.

We use five noninterchangeable labels.

PROVED an analytic theorem in the exact scope of its source;

CERTIFIED	a finite exact, stored, or outward computation;
CONDITIONAL	a valid implication with unproved physical inputs;
SCOPED NEGATIVE	a theorem rejecting one named information class, model, or method, not every physical realization;
OPEN/NOT TESTABLE	a required source, map, clock, uniformity, or observation is absent.

Three firewalls govern every cross-batch statement. First, actual noisy, synthetic spectral, deterministic combinatorial, and abstract coefficient objects retain their types. Second, fixed order, a selected slow diagonal, and the physical logarithmic first-alias clock are not interchangeable. Third, normalized or selected-window control is not an unnormalized complete prefix theorem.

2 Executable provenance theorem

The range has eight review anchors:

291, 301, 311, 321, 331, 341, 351, 361.

They close the consecutive ten-paper blocks beginning at RH-282. Four numbers, RH-302, RH-303, RH-304, and RH-306, have an additional empty legacy directory. The audit chooses the candidate containing a README, TeX source, and PDF and records every alternative instead of deleting it.

Theorem 2.1 (Volume IV provenance theorem). *The executable inventory defines a canonical nonempty source directory for each integer $282 \leq n \leq 361$. The resulting map contains eighty consecutive labels, records exactly four duplicate-directory groups, and assigns every label to exactly one of eight ten-paper phases. Selected README, manuscript, theorem-ledger, roadmap, and result files are bound by SHA-256.*

Proof. Directory discovery uses the numerical prefix rather than a hand-written list. For each label the canonical candidate maximizes the required-file score and then the finite file count; the chosen candidate is rejected unless all three publication anchors are present. The expected integer list and the union of the declared phase intervals are both checked against $\{282, \dots, 361\}$. Hash replay is performed independently by the archive verifier. This proves presence and identity, not the mathematical claims in the source files. $\square$

3 The common coefficient dictionary

On the source-locked Hardy clock

$$k = \frac{\log(1/\sigma)}{2\log \lambda} + O(1), \quad 2 \leq n < 4k,$$

let $p_{\sigma,n}$ denote the actual direct coefficient, $q_{\sigma,n}$ the actual full-trace coefficient, $h_{\sigma,n}$ the actual modulus-complete Hardy head, and $s_{k,n}$ the deterministic graded counterloop. Write $d_{\sigma,k,n} = h_{\sigma,n} - s_{k,n}$. In the typed source formulas,

$$p = \tau - a = q - d, \quad d = h - s, \quad qqquad q = p + d, \quad qqquad h = s + d. \quad (1)$$

Equation (1) is algebra, not a spectral identification. In particular, it does not say that the roots or moments defining s occur inside the noisy operator, that the rank of the deterministic

model is a physical rank, or that the off-cloud coefficient q is controlled by p without a bound for d .

The natural unnormalized defect budget is

$$D_{4k}(R) = \sum_{2 \leq n < 4k} |h_{\sigma,n} - s_{k,n}| \frac{R^n}{n}. \quad (2)$$

It is distinct from the normalized upper-band error, from coordinatewise relative matching, and from any maximum over a selected window.

4 RH-282–RH-291: noisy heads and determinant tails

This phase reopens the operator route without requiring a global moving Riesz projector. RH-282 constructs an actual modulus-complete normal spectral tail in the data type required by an earlier conditional tail criterion. RH-283 gives the sharp logarithmic block clock for the declared mass-and-cap class. RH-284 defines the canonical minimal head relative to a threshold; the word “canonical” is relative to that threshold and does not mean a canonical Hilbert–Polya spectrum.

RH-285 proves uniform fixed-derivative bounds for the complementary $\det_2$ tail. RH-287 obtains a synchronized growing actual-noisy prefix on a selected, nonquantitative diagonal. RH-288 gives the exact prefix–tail gluing criterion and separates two error types. The direct complement prefix remains open. RH-289 constructs hidden-shell moment nonidentifiability, and RH-290 records why two typed ledgers cannot be united coordinatewise.

The durable gain is a complete analytic tail and a real, finite actual head. The durable limitation is rate: no result places the complete direct prefix on the logarithmic first-alias clock. A slow diagonal is an existence statement, not an admissible replacement for the required clock.

5 RH-292–RH-301: weighted prefixes and clocks

The weighted norm magnifies high orders, so seemingly strong coordinatewise information may be insufficient. RH-292 shows how analytic tail absorption shortens a bridge clock. RH-293 obtains a sharp coefficient-error exponent for its information class. RH-294 proves a weighted full-trace bridge along a rate-free selected clock, while RH-295 gives a scoped counterexample: the unweighted maximum can vanish on the correct window even as the weighted prefix diverges.

RH-296 proves that the existing orbit-localization and analytic tail clock ranges are disjoint for the named method. RH-297 counts the one or two exact counterloop aliases crossed by the natural clock. RH-298 gives a square-root parity barrier for separate absolute majorants, leaving grouped signed cancellation open. RH-299 proves a zero-padded root- ℓ^1 transport criterion and its sharp rate, but actual matching is absent. RH-300 gives strict outer-circle H^∞/H^2 direct-prefix criteria and an endpoint counterexample.

These are genuine theorems about rates, clocks, and information classes. They do not supply the actual outer-circle norm required to activate the criteria. Finite coefficient rows cannot replace that missing uniform analytic input.

6 RH-302–RH-311: annular mass and Hardy barriers

RH-302 reduces an annular tail problem to a moving head. RH-303 proves that fixed-order head-moment transport is necessary. RH-304 computes the mass exponent demanded by minimal-clock relative matching, while RH-305 gives an interior annular mass-rate lower barrier. RH-306 is sharp for the declared coefficient-envelope class; sharpness in that class is not a theorem about the actual noisy spectrum.

RH-307 treats shrinking-annulus thresholds, RH-308 identifies the optimal Hardy conversion order, and RH-309 separates endpoint H^2 membership from H^∞ exclusion and convergence. RH-310 aligns alias, parity, and localization clocks and records a typed matching necessity. The RH-311 review shows that the strongest rows in the two branches each miss a different leaf; their scores cannot be joined into a complete row.

The phase therefore isolates the exact annular price of a physical bridge. It does not prove endpoint convergence or weighted cross-branch glue.

7 RH-312–RH-321: endpoint expansions and synthetic sharpness

RH-312 separates a universal endpoint logarithm from an analytic remainder. RH-313 resolves parity-orthogonal endpoint channels, and RH-314 proves the optimal polynomial endpoint rate $N^{-1/2}$ in its declared setting. RH-315 constructs exact root-of-unity moment packets. RH-316 realizes exact finite integer spectral prefixes, RH-317 proves sharp rank and squared-mass growth, and RH-318 identifies the optimal endpoint mass law and constant. RH-319 achieves genuine strict-annulus saturation. RH-320 shows that an escaping synthetic packet can obstruct an endpoint-energy route.

The word “spectral” in this block describes an explicit finite or growing synthetic spectral class. No source identifies this class with the spectrum of the physical folded Gaussian operator. Consequently the sharp synthetic laws are valuable extremizers and countermodels, but they cannot promote an actual endpoint theorem.

8 RH-322–RH-331: local first-alias interfaces

The first-alias block begins with actual boundary geometry. RH-322 proves a folded-row half-line profile and phase stability. RH-323 retains orientation in an affine Gaussian chain and proves an exact retained-coordinate L^1 isometry. RH-324 replaces the first tangent leg by the physical row and finds a sharp $O(\sigma)$ curvature remainder.

RH-325 separates Markov path composition from operator trace composition. For path laws, transported row errors compose by contraction. For traces, a separately bounded observation and controlled prefix/suffix products are necessary; a dimension-free Markov-to-trace implication is false. Its stability window is

$$\gamma < 1 - \frac{\log(1.4)}{\log \lambda} = 0.3503698834605293 \dots,$$

provided every local operator error is $O(\sigma)$ and the required weights are actual upper bounds.

RH-326 gives the Hardy-normalized parity and first-alias packet identity. RH-327–RH-330 develop localized trace ledgers, matching equations, exchange models, and a grouped signed transfer criterion. Their negative examples are scoped to the named information classes or frozen models. RH-331 ends at an exact transfer criterion whose actual replacement hypotheses are open.

9 RH-332–RH-341: actual observation and replacement

RH-332 supplies the second sharp physical affine-leg remainder, but no single global row theorem follows. RH-333 proves escape of the canonical raw full-line forward affine retained path on the first-alias clock. It explicitly does not refute a cyclic bridge, Doob transform, compact folded reference, or nonlinear closing profile.

RH-334 repairs the physical observation: multiplication marks the cyclic basepoint of the backward folded operator, and the signed/folded localized traces agree for powers at least two. A forward probability law is not thereby identified with that trace. RH-335 localizes the parity projector as a signed measure and proves nonuniqueness of cellwise extensions; projector mass need not be a probability.

RH-336–RH-340 refine boundary atoms, signed allocations, and actual/model replacement formulas. The RH-341 review records a decisive distinction. Separate absolute orbit, diffuse, and head majorants encounter a divergent two-atom burden, while grouped signed cancellation remains unknown. Thus the separate-majorant route is stopped, but the physical aggregate is not refuted.

10 RH-342–RH-351: boundary atoms and lower completion

RH-342 freezes the common Hardy head and deterministic counterloop data types. Subsequent papers decompose critical and lower-sideband boundary orbits, track parity phases, and prove exact algebraic identities. RH-345 and RH-347 expose scalar balance underdetermination; RH-349–RH-350 prove phase incompatibility for named lower-sideband completion mechanisms.

RH-351 then separates a deterministic P/S ledger from the unknown actual signed remainder. Opposite abstract completions can satisfy the same visible premises, showing that those premises do not determine the desired selected physical verdict. This is an information-class theorem. It does not assert that both completions are realized by noisy operators.

Odd orders, upper aliases, the complete E_{off} aggregate, and the RH-241 moving noisy all-order envelope all remain open at this stage.

11 RH-352–RH-361: actual tails and deterministic counterloops

The final phase contains two real advances of different types. RH-352 proves an actual growing lower-even normalized result for p and a selected signed quantity. RH-353 obtains a phase-free two-coordinate gap for an actual critical/first-lower normalized quantity. RH-354 proves an actual parity-free normalized direct tail above a moving cut. These statements are not an unnormalized prefix including low orders.

RH-355–RH-360 then solve a deterministic counterloop problem: the complete strict-upper burden, mesoscopic crossover, linear-depth profile, geometric terminal-lag localization, logarithmic inverse-accuracy law, and an exponential-tilt phase diagram. Every unconditional statement here is for s . Transfer to the actual head is explicitly conditional on $D_{4k}(R) \rightarrow 0$.

RH-361 closes the batch with an exact finite coefficient-fiber theorem and a ten-layer audit. It proves that current data do not determine a uniform q or h budget in the abstract coefficient class. It does not construct a physical pair with arbitrary defect.

12 Typed nonpromotion theorem

The synthesis can now state the cross-batch inference exactly.

Theorem 12.1 (Typed nonpromotion theorem). *Fix the common clock and orders $2 \leq n < 4k$. The proved Volume IV inputs consisting of*

- (i) *selected or normalized actual statements for p and related signed quantities, and*
- (ii) *unconditional deterministic statements for s*

do not, by the identities (1) alone, imply an unnormalized weighted bound for the complete actual q or h . Such an implication requires an independent same-clock estimate for $d = h - s$ on the complete order range.

Proof. Both target identities contain the unpaid term: $q = p + d$ and $h = s + d$. Neither a selected/normalized estimate for p nor a deterministic estimate for s controls the weighted norm of d . Moreover, normalization cannot be removed without its scale, and a moving upper window does not include the omitted low orders. Hence the claimed full budgets do not follow from the stated premises. This is a theorem about implication and types; it makes no assertion about which defects are physically realizable. $\square$

For completeness, on a fixed finite set I and positive weights, hold signed arrays p, s fixed and let e range over all signed arrays. Define

$$d[e] = e, \quad q[e] = p + e, \quad h[e] = s + e.$$

Along $e = tv$ with $v \neq 0$, the weighted norms of $q[e]$ and $h[e]$ are unbounded as $|t| \rightarrow \infty$. Choosing $e = -p$ or $e = -s$ makes the respective norm zero. This verifies information-class nonidentifiability, but the two zero choices need not coincide and arbitrary e is not declared physical.

13 Frontier theorem and reopening tests

Theorem 13.1 (Source-relative frontier theorem). *Relative to the canonical RH-282–RH-361 sources, no proved result supplies (2), a genuine physical obstruction to it, a complete actual q/E_{off} theorem on the same clock, or the RH-241 moving noisy all-order envelope with its coefficient bridge. Therefore the source-relative verdict is `NOT_TESTABLE`, and RH-362 is not activated by this synthesis.*

Proof. Each ten-layer review records its first missing typed leaf. RH-291 and RH-301 stop at a rate-free or inactive prefix criterion; RH-311 lacks cross-branch glue; RH-321 is sharp only for a synthetic class; RH-331 and RH-341 lack all-leg actual trace replacement; RH-351 lacks the actual growing signed remainder; and RH-361 explicitly records (2) as an assumption of every actual-head inheritance statement. The executable audit binds these reviews and their source ledgers. Absence of a source is not a physical impossibility theorem, so the conclusion is `NOT_TESTABLE` rather than a global negative. $\square$

One new direction survives the scoped raw-forward obstruction. For full-line affine Gaussian rows whose composition has slope A , shift B , and variance V , the cyclic diagonal integral is

$$\int_{\mathbb{R}} \frac{1}{\sqrt{2\pi V}} \exp\left(-\frac{((1-A)x - B)^2}{2V}\right) dx = \frac{1}{|1-A|}, \quad A \neq 1.$$

Raw forward variance growth therefore does not by itself obstruct a cyclic diagonal functional. However the full-line kernel is not Hilbert–Schmidt on $L^2(\mathbb{R})$, and compact truncation and inverse row normalizers act in opposite directions. A physical reopening must first define the complete signed/folded compact reference in the RH-334 observation type and test

$$C_{\sigma,k} = 1 + o((\beta R)^{-2k}).$$

Only after this correction-factor test may all-leg curvature Duhamel and observation-weight estimates be attempted.

14 Reproduction protocol and limitations

The executable protocol has four layers:

1. discover and canonicalize the eighty numerical labels;

2. bind selected source files and the eight review anchors by SHA-256;
3. test the range, aliases, phase cover, typed identities, route coordinate, Gate values, and forbidden claims;
4. build the PDF, archive local publication artifacts, and independently replay local and external hashes.

This protocol detects omission, alias confusion, source drift, and explicit claim promotion. It does not re-prove eighty papers, certify an asymptotic from finite data, or decide whether a currently undefined physical bridge exists. Numerical tables in the sources retain exactly their original status.

15 Claim boundary

Volume IV proves Theorem 2.1, the typed logical statement in Theorem 12.1, and the source-relative audit conclusion in Theorem 13.1. It preserves all stronger mathematical claims in their atomic sources and does not silently strengthen them.

Gates A–E remain false/open. The volume does not construct a Hilbert–Polya operator, identify Riemann zeros, prove a von Mangoldt trace, prove equality with the completed-zeta divisor, or prove the Riemann Hypothesis. The active coordinate remains

`actual_same_clock_unnormalized_head_transport_open`.

A Atomic source index

This machine-generated table lists the canonical directory for every numbered source in Volume IV and resolves the four legacy alias groups. It is a provenance locator, not a replacement for the source theorem ledgers.

source	canonical directory
RH-282	RH-282-modulus-complete-spectral-tail-certificate
RH-283	RH-283-sharp-logarithmic-spectral-block-clock
RH-284	RH-284-canonical-modulus-complete-spectral-head
RH-285	RH-285-uniform-det2-tail-derivative-envelope
RH-286	RH-286-finite-radius-monodromy-cloud-centering
RH-287	RH-287-synchronized-growing-prefix-counterloop-bridge
RH-288	RH-288-prefix-tail-determinant-gluings-criterion
RH-289	RH-289-finite-moment-shell-nonidentifiability
RH-290	RH-290-triple-branch-spectral-counterloop-ledger
RH-291	RH-291-ten-layer-spectral-tail-frontier-review
RH-292	RH-292-tail-absorbed-weighted-bridge-clock
RH-293	RH-293-sharp-minimal-clock-prefix-rate
RH-294	RH-294-rate-free-weighted-full-trace-diagonal-bridge
RH-295	RH-295-exact-clock-unweighted-prefix-obstruction
RH-296	RH-296-tail-localization-clock-separation
RH-297	RH-297-natural-counterloop-alias-ledger
RH-298	RH-298-square-root-parity-majorant-barrier
RH-299	RH-299-zero-padded-root-l1-bridge-clock
RH-300	RH-300-annular-analytic-prefix-criteria
RH-301	RH-301-ten-layer-weighted-prefix-frontier-review

source	canonical directory
RH-302	RH-302-annular-tail-moving-head-reduction
RH-303	RH-303-annular-fixed-order-head-transport-necessity
RH-304	RH-304-minimal-clock-complement-mass-demand
RH-305	RH-305-interior-annular-norm-rate-barrier
RH-306	RH-306-sharp-annular-coefficient-envelope-saturation
RH-307	RH-307-critical-shrinking-annulus-tail-threshold
RH-308	RH-308-hardy-dominant-annular-coefficient-conversion
RH-309	RH-309-endpoint-hardy-mismatch-barrier
RH-310	RH-310-critical-alias-parity-localization-coincidence
RH-311	RH-311-ten-layer-annular-mass-frontier-review
RH-312	RH-312-endpoint-logarithmic-singularity-decomposition
RH-313	RH-313-parity-orthogonal-endpoint-hardy-splitting
RH-314	RH-314-optimal-endpoint-logarithm-polynomial-approximation
RH-315	RH-315-root-of-unity-moment-packet-isolation
RH-316	RH-316-recursive-integer-spectral-prefix-realization
RH-317	RH-317-sharp-spectral-prefix-rank-mass-law
RH-318	RH-318-optimal-endpoint-spectral-mass-approximation
RH-319	RH-319-genuine-spectral-annular-envelope-saturation
RH-320	RH-320-escaping-packet-endpoint-energy-obstruction
RH-321	RH-321-ten-layer-endpoint-spectral-frontier-review
RH-322	RH-322-certified-critical-folded-row-half-line-profile
RH-323	RH-323-oriented-paired-affine-gaussian-chain
RH-324	RH-324-sharp-physical-endpoint-affine-leg-remainder
RH-325	RH-325-moving-order-duhamel-composition-criterion
RH-326	RH-326-parity-renormalized-first-alias-packet-identity
RH-327	RH-327-neighboring-shell-coupling-cancellation-budget
RH-328	RH-328-joint-alias-parity-shell-matching-equation
RH-329	RH-329-validated-isolated-exchange-model-audit
RH-330	RH-330-joint-cancellation-full-trace-transfer-criterion
RH-331	RH-331-ten-layer-first-alias-frontier-review
RH-332	RH-332-sharp-physical-repelling-return-affine-leg-remainder
RH-333	RH-333-raw-forward-affine-tube-escape-obstruction
RH-334	RH-334-gauge-fixed-physical-first-alias-observation-map
RH-335	RH-335-projector-localized-parity-ledger-and-cellwise- extension-nonuniqueness
RH-336	RH-336-projector-mass-first-alias-threshold-and-isospectral- cell-obstruction
RH-337	RH-337-algebraic-clock-drift-and-parity-alias-replacement- obstruction
RH-338	RH-338-boundary-orbit-far-atom-and-signed-diffuse- compensation-obstruction
RH-339	RH-339-first-lower-sideband-orbit-atom-compensation- obstruction
RH-340	RH-340-synchronized-determinant-prefix-and-two-order-orbit- head-compensation-obstruction
RH-341	RH-341-ten-layer-actual-first-alias-replacement-frontier- review
RH-342	RH-342-common-hardy-head-counterloop-rank-lock-obstruction
RH-343	RH-343-equal-rank-equal-mass-first-alias-underdetermination
RH-344	RH-344-complete-critical-boundary-orbit-atom-decomposition

source	canonical directory
RH-345	RH-345-double-alias-parity-phase-compensation-obstruction
RH-346	RH-346-complete-lower-sideband-boundary-orbit-decomposition
RH-347	RH-347-lower-sideband-scalar-balance-underdetermination
RH-348	RH-348-punctured-lower-even-boundary-orbit-ladder
RH-349	RH-349-two-lower-sideband-phase-incompatibility
RH-350	RH-350-growing-depth-lower-sideband-phase-incompatibility
RH-351	RH-351-ten-layer-signed-completion-frontier-review
RH-352	RH-352-modulus-cap-forced-growing-ladder-signed-cancellation
RH-353	RH-353-critical-first-lower-actual-signed-completion-gap
RH-354	RH-354-parity-free-near-alias-direct-tail-envelope
RH-355	RH-355-upper-alias-counterloop-burden-and-head-transfer-precision
RH-356	RH-356-sharp-post-first-alias-mesoscopic-crossover
RH-357	RH-357-uniform-linear-depth-upper-counterloop-profile
RH-358	RH-358-terminal-lag-geometric-localization
RH-359	RH-359-logarithmic-terminal-window-accuracy-thresholds
RH-360	RH-360-terminal-lag-exponential-tilt-phase-transition
RH-361	RH-361-ten-layer-signed-completion-and-upper-counterloop-review
